Problem 3.18

Consider the amount function $A(t) = 12(1.01)^{4t}$.

(a) Find the principal.

(b) Find the effective annual interest rate.

Solution

(a) The principal is the value at time $t = 0$:

$$A(0) = 12(1.01)^{4(0)}.$$

Hence,

$$A(0) = 12.$$

(b) First compute the amount at time $t = 1$:

$$A(1) = 12(1.01)^4.$$

Therefore,

$$A(1) = 12.48724.$$

The effective annual interest rate is

$$i = \frac{A(1) - A(0)}{A(0)}.$$

Thus,

$$i = \frac{12.48724 - 12}{12} = 0.04060.$$

Therefore,

$$i = 4.06\%.$$